

Optimization of Graph Squares

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Abstract

In this paper, we investigate the matrix properties of graph squares, focusing in particular on the relationship between the adjacency matrix of a graph and the corresponding adjacency of its square.

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1 Introduction

The problem of finding the square root of a graph is a well-studied field in both mathematics and computer-science, with specific relation to distributed computing, network design and computational biology. Research in this area has primarily focused around the undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two.

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the complexity of reconstructing H from G . In contrast, rather

than studying the construction problem directly, we analyze the relationship between the adjacency matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program.

We then examine the relaxed linear program which can be evaluated in polynomial time using interior point method, and integrate existing literature on special root structures as a stepping-stone towards generating a true polynomial time algorithm.

2 Background

Let the *graph root* $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. Denote by $d_H(u, v)$ the *graph distance* defined as the number of edges of the shortest path between vertices u and v in H . If no such path exists, $d_H(u, v) = \infty$.

The *square* of the graph H , denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H . This is the standard definition of the square of a graph.

Using an alternative construction, let the *graph square* $G = (V, E_G)$ be an undirected graph defined on the same vertex set V . Let $A_H \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of H , where

$$(A_H)_{ij} = \begin{cases} 1, & \text{if } ij \in E_H, \\ 0, & \text{otherwise,} \end{cases}$$

and let $D_H \in \mathbb{R}^{n \times n}$ denote the degree matrix of H , where

$$(D_H)_{ij} = \begin{cases} \deg_H(i), & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (2)$$

Lemma 1. *The graph G defined by (2) is equivalent to the graph square H^2 defined in (1)*

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (3)$$

Proof. For $i \neq j$, since D_H is diagonal, have $(D_H)_{ij} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ij} = (A_H)_{ij}^2 + (A_H)_{ij}.$$

($\Rightarrow$) Suppose $ij \in E_G$ such that $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$. This implies that at least one of $(A_H)_{ij} = 1$ or $(A_H)_{ij}^2 \geq 1$ must be true. If $(A_H)_{ij} = 1$, then $d_H(i, j) = 1$. If $(A_H)_{ij}^2 \geq 1$, then there exists

a vertex k such that $ik, kj \in E_H$, and hence $d_H(i, j) \leq 2$. Therefore $1 \leq d_H(i, j) \leq 2$, and thus $ij \in E_{H^2}$.

($\Leftarrow$) Suppose $ij \in E_{H^2}$ such that $1 \leq d_H(i, j) \leq 2$. D_H is diagonal such that $(D_H)_{ij} = 0$. If $d_H(i, j) = 1$, then $(A_H)_{ij} = 1$. If $d_H(i, j) = 2$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $(A_H)_{ij}^2 \geq 1$. In either case, $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$, so $(A_G)_{ij} = 1$, and hence $ij \in E_G$.

For $i = j$, since all graphs are simple, have $(E)_{ii} = 0$, $(A)_{ii} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ii} = (A_H)_{ii}^2 - (D_H)_{ii}.$$

$(A_H)_{ii}^2 - (D_H)_{ii} = 0$ is always true since the degree matrix of H always has diagonal values $(D_H)_{ii} = \sum_k x_{ik}$ and each edge from vertex i to its neighbour in H will always create a loop

to itself in the square such that $(A_H)_{ii}^2 = \sum_{k=1} x_{ik} x_{ki} = \sum_{k=1} x_{ik}^2$ by the symmetry of adjacency matrix A_H . Since $x_{ik} \in \{0, 1\}$, the square of the boolean is identical to the boolean x_{ik} such that $(A_H)_{ii}^2 = \sum_k x_{ik} = (D_H)_{ii}$ and $(A_G)_{ii} = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H)$ as required.

Combining both gives that $E_G = E_{H^2}$, and so $G = H^2$. □

The formulation of the adjacency matrix of the graph square in Equation (2) is very powerful. However, the presence of the indicator function introduces a nonlinearity that prevents the problem from being expressed as a purely quadratic form, thereby limiting the direct applicability of existing results from the quadratic optimization literature.

Lemma 2. *If there are no K_3 triangle subgraphs or any C_4 square subgraphs in the graph root such that H has girth of length 5, then Equation (2) can be simplified to become*

$$A_G = ((A_H)^2 - D_H + A_H). \quad (4)$$

Proof. Since the graph root has girth of length 5 there are no cycles of length 3 or 4 in H . Thus for every pair of vertices $i \neq j \in V$ there is either no cycle that connects i, j or there is a cycle of at least length 5 between them. In all cases $(D_H)_{ij} = 0$. If there is no cycle, then there is at most one path between i, j

- i) If the path is of length 1, then $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If the path is of length 2, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the path is of length ≥ 3 or a path does not exist, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If there is a cycle of length ≥ 5 between i, j such that there are two paths between i, j

- i) If one path is of length 1, then the other is of length ≥ 4 and hence $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If one path is of length 2, then the other is of length ≥ 3 and hence $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the paths are both of length ≥ 3 , then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If $i = j$, then $(A_H)_{ii} = 0$, $(D_H)_{ii} = \deg_H(i)$, and $(A_H)_{ii}^2 = \deg_H(i)$. Thus,

$$(A_H)_{ii}^2 - (D_H)_{ii} + (A_H)_{ii} = \deg_H(i) - \deg_H(i) + 0 = 0 = (A_G)_{ii}.$$

□

2.1 Integer Program

Let $x_{ij} \in \{0, 1\}$ be the decision variables of A_H , let $y_{ikj} \in \{0, 1\}$ be the decision variable for $x_{ik}x_{kj}$ in $(A_H)^2_{ij}$ and let $z_{ij} \in \mathbb{Z}_{\geq 0}$ represent the argument inside the indicator function.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix} \text{ such that}$$

$$(A_H)^2 = \begin{bmatrix} \sum_{k=1}^n x_{1k}x_{k1} & \cdots & \sum_{k=1}^n x_{1k}x_{kj} & \cdots & \sum_{k=1}^n x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{ik}x_{k1} & \cdots & \sum_{k=1}^n x_{ik}x_{kj} & \cdots & \sum_{k=1}^n x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{nk}x_{k1} & \cdots & \sum_{k=1}^n x_{nk}x_{kj} & \cdots & \sum_{k=1}^n x_{nk}x_{kn} \end{bmatrix}.$$

Then with known constants $g_{ij} \in \{0, 1\}$ from A_G , can reformulating (2) as an integer program by equating entries of each matrix to give the following linear system of n^2 equations.

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{k1} - (D_H)_{11} + x_{11}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\right) \end{bmatrix}. \quad (5)$$

The matrix D_H is diagonal can be represented as $(D_H)_{ii} = \sum_k x_{ik}$ where $(D_H)_{ij} = 0$ for $i \neq j$. Let

$M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. The known z_{ij} represent the terms inside the indicator function such that

$$g_{ij} = 1 \iff z_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff z_{ij} = 0.$$

The above condition can be modelled using a standard big- M formulation, and the resulting integer program is given by

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (7)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (8)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (9)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (10)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (12)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (13)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (14)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (15)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (16)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}. \quad (17)$$

In addition, the adjacency matrix x must satisfy further constraints arising from symmetry and diagonality. Although analogous constraints could also be imposed on the auxiliary variables z they are omitted due to redundancy and potentially introduces artificial degeneracy in the linear program.

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (18)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (19)$$

Proposition 1. *Let G be a graph and consider the integer program whose variables $x_{ij} \in \{0, 1\}$ encode a graph H with adjacency matrix $A_H = (x_{ij})$. Then G has a square root H if and only if the integer program has a feasible solution x such that the corresponding graph H satisfies $H^2 = G$.*

($\Rightarrow$) Suppose G has a square root H and let A_H be the adjacency matrix of H . $x_{ij} = (A_H)_{ij}$. Construct the variables x_{ij}, y_{ikj}, z_{ij} and the constant g_{ij} as such

i) Let $x_{ij} = 1$ if $ij \in E_H$ and $= 0$ otherwise.

ii) Let $y_{ikj} = x_{ik}x_{kj}$ for all i, k, j .

iii) Let $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}$ for all i, j .

iv) Let $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik} = 0$ since $x_{ij} \in \{0, 1\} \implies x_{ik}^2 = x_{ik}$.

v) Let $g_{ij} = 1$ if $ij \in E_G$ and $= 0$ otherwise.

Verifying constraints of the integer program for the variables

- i) As $x_{ij} \in \{0, 1\} \implies y_{ikj} \in \{0, 1\}$, Equations (6)-(8) and Equations (16)-(17) are satisfied.
- ii) As H is a simple directed graph, its adjacency matrix is hollow and symmetric such that Equations (18)-(19) are satisfied.
- iii) As $y_{ikj} = x_{ik}x_{kj}$ and all variables are binary, Equations (13)-(15) are satisfied.
- iv) Equation (10) by definition is satisfied.

v) As $z_{ii} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik}$, Equation (9) is satisfied.

vi) As $z_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$ count the number of walks of length at most 2 between

$$g_{ij} = 1 \implies z_{ij} \geq 1 \text{ and } g_{ij} = 0 \implies z_{ij} = 0$$

($\Leftarrow$) Suppose the integer program on graph G has a feasible solution x , and let x_{ij} define a graph H by

$$ij \in E_H \iff x_{ij} = 1.$$

As x_{ij} satisfies the constraints of the integer program,

- i) Equation (6) and Equation (16) gives $x_{ij} \in \{0, 1\}$ for all i, j ,
- ii) Equation (18) and Equation (19) give $x_{ij} = x_{ji}$ and $x_{ii} = 0$,

such that x_{ij} defines a hollow, symmetric matrix with all elements being either 0 or 1 which is a valid adjacency matrix of a graph. Examining the auxiliary variables y_{ikj} and z_{ij} ,

- i) Equations (13)-(15) give the strict equality $y_{ikj} = x_{ik}x_{kj}$ for all i, j, k ,
- ii) Equation (10) gives $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$ for all $i \neq j$,
- iii) Equation (9) gives $z_{ii} = \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik} = 0$ for all i since

$$\sum_{k=1}^n x_{ik}x_{ki} = \sum_{k=1}^n x_{ik}^2 = \sum_{k=1}^n x_{ik},$$

by symmetry and since $x_{ij} \in \{0, 1\}$.

The indicator constraints Equations (11)-(12) thus give for $i \neq j$

$$g_{ij} = 1 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} = 0.$$

Since $x_{ij} \in \{0, 1\}$, this is equivalent to

$$g_{ij} = 1 \iff (x_{ij} = 1) \vee (\exists k \text{ such that } x_{ik} = 1 \wedge x_{kj} = 1) \text{ and } g_{ij} = 0 \iff x_{ij} = 0.$$

By definition of H this means that $g_{ij} = 1 \iff (ij \in E_H) \vee (\exists k \text{ such that } ik, kj \in E_H)$ which is equivalent to

$$g_{ij} = 1 \iff 1 \leq d_H(i, j) \leq 2.$$

Now apply Lemma 1 gives $ij \in E_G \iff ij \in E_H^2$ such that $G = H^2$.

2.2 Linear Program

The linear relaxation of the integer program defined by Equations (6)–(19) is constructed in order to generate a feasible weighted solution in polynomial time algorithm. It is done with identical constraints, however the decision variables (6), (7) and (8) are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (20)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (21)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (22)$$

The results from the linear programming can give *integer solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist such that the results in Figure 1 can be generated.

However, the direct linear relaxation needs additional constraints after the linear relaxation. Consider $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} + x_{ii}$ which can be simplified to $z_{ii} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik}$ and is equal to 0 when x is binary.

In the linear relaxation this no longer holds for fractional values of x_{ik} (IS THIS REALLY TRUE? NOT ADDING CONSTRAINT STILL GIVES ALL VALID SOLUTIONS?)

Proposition 2.

3 Linear Program

3.1 Max Objective Function

Consider the addition of an objective function maximizing the total weights on all edges to the linear relaxation defined by Equations (9)–(22) in Section 2.2

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (23)$$

This addition gives the improved results in Figure 2.

3.2 Singular Triangular Decomposition

Using the local triangular structure of the graph, certain edge assignments in a square root can be fixed. For a vertex $s \in V$, define

$$T(s) = \{ \{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E \}$$

to be the set of triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours are structurally distinct with different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the leaf-adjacent vertex in the square root. We enforce the deterministic assignment

$$x_{sw} = 1, x_{ws} = 1 \text{ where } w = \arg \min\{\deg_G(u), \deg_G(v)\}. \quad (24)$$

If $\deg_G(u) = \deg_G(v)$, no constraint is imposed. This addition gives the improved results in Figure 3 whilst maintaining polynomial runtime.

In the case of $\deg_G(u) = \deg_G(v)$, enforcement is done by arbitrarily choosing one of the edges of only one such triangle. This is done recursively until all such singular triangles are resolved. This gives the improved results in Figure 4.

Ross and Harary in 1960 gave an algorithm for constructing a tree root by using characterizations of the square graph to conclude that existing tree square roots are unique up to isomorphism [RH60].

Proposition 3. *Singular triangular decomposition completely categorizes all triangle roots in polynomial time linear program.*

3.3 Rounding

LP fractional solutions are derived from the LP when the root H exists and does not exist. An attempt for rounding is achieved such that the classification in Figure 5 is obtained.

3.4 Dual Problem

We now consider the dual formulation of the max objective linear program defined in Section 3.1 by first restating it in canonical form.

Objective Function:

$$\min - \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (25)$$

Variables:

$$x_{ij} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, j \in \{1, \dots, n\}, \quad (26)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, k, j \in \{1, \dots, n\}, \quad (27)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (28)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} = 0, \forall i \in \{1, \dots, n\}, \quad (29)$$

$$z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} = 0, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (30)$$

$$(2) \text{ Indicator function: } z_{ij} - M g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (31)$$

$$-z_{ij} + g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (32)$$

$$(3) \text{ Product: } y_{ikj} - x_{ik} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (33)$$

$$y_{ikj} - x_{kj} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (34)$$

$$x_{ik} + x_{kj} - y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (35)$$

$$(4) \text{ Symmetry: } x_{ij} - x_{ji} = 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (36)$$

$$(5) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (37)$$

$$(6) \text{ Variable Bound: } x_{ij} - 1 \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (38)$$

$$y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}. \quad (39)$$

This gives the dual problem with

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n \left(-M g_{ij} \gamma_{ij} + g_{ij} \delta_{ij} - \lambda_{ij} \right) - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n (\eta_{ikj} + \mu_{ikj}). \quad (40)$$

Dual Variables:

$$\alpha_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (29)}), \quad (41)$$

$$\beta_{ij} \in \mathbb{R}, \quad \forall i \neq j \in \{1, \dots, n\}, (\text{associated with (30)}), \quad (42)$$

$$\gamma_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (31)}), \quad (43)$$

$$\delta_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (32)}), \quad (44)$$

$$\epsilon_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (33)}), \quad (45)$$

$$\zeta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (34)}), \quad (46)$$

$$\eta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (35)}), \quad (47)$$

$$\theta_{ij} \in \mathbb{R}, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (36)}), \quad (48)$$

$$\iota_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (37)}), \quad (49)$$

$$\lambda_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (38)}), \quad (50)$$

$$\mu_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, (\text{associated with (39)}). \quad (51)$$

Dual Constraints:

$\forall i, j \in \{1, \dots, n\}$, each primal x_{ij} :

$$\begin{cases} -1 - \sum_{k=1}^n \epsilon_{iik} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iik} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \geq 0, & \text{if } j = i, \\ -1 + \alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \theta_{ij} - \theta_{ji} + \lambda_{ij} \geq 0, & \text{if } j \neq i, \end{cases} \quad (52)$$

$\forall i, k, j \in \{1, \dots, n\}$, each primal y_{ikj} :

$$\begin{cases} -\alpha_i + \epsilon_{iki} + \zeta_{iki} - \eta_{iki} + \mu_{iki} \geq 0, & \text{if } j = i, \\ -\beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \geq 0, & \text{if } j \neq i, \end{cases} \quad (53)$$

$\forall i, j \in \{1, \dots, n\}$, each primal z_{ij} :

$$\begin{cases} \alpha_i + \gamma_{ii} - \delta_{ii} = 0, & \text{if } j = i, \\ \beta_{ij} + \gamma_{ij} - \delta_{ij} = 0, & \text{if } j \neq i. \end{cases} \quad (54)$$

whereby the Lagrangian is

$$\begin{aligned} \mathcal{L}(x, y, z) = & - \sum_{i=1}^n \sum_{j=1}^n x_{ij} + \sum_{i=1}^n \alpha_i \left(z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} \right) + \sum_{i \neq j} \beta_{ij} \left(z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} \right) \\ & + \sum_{i=1}^n \sum_{j=1}^n \gamma_{ij} (z_{ij} - M g_{ij}) + \sum_{i=1}^n \sum_{j=1}^n \delta_{ij} (-z_{ij} + g_{ij}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \epsilon_{ikj} (y_{ikj} - x_{ik}) \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \zeta_{ikj} (y_{ikj} - x_{kj}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} (x_{ik} + x_{kj} - y_{ikj} - 1) \\ & + \sum_{i=1}^n \sum_{j=1}^n \theta_{ij} (x_{ij} - x_{ji}) + \sum_{i=1}^n \iota_i x_{ii} + \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} (x_{ij} - 1) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj} (y_{ikj} - 1). \end{aligned}$$

Collecting terms gives

$$\begin{aligned} \mathcal{L}(x, y, z) = & \sum_{i=1}^n \sum_{j=1}^n x_{ij} \left[-1 + \mathbf{1}_{\{j=i\}} \left(\alpha_i - \sum_{k=1}^n \epsilon_{iki} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iki} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \right) \right. \\ & \left. + \mathbf{1}_{\{j \neq i\}} \left(\alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \lambda_{ij} \right) + \theta_{ij} - \theta_{ji} \right] \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n y_{ikj} \left[-\mathbf{1}_{\{j=i\}} \alpha_i - \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \right] \\ & + \sum_{i=1}^n \sum_{j=1}^n z_{ij} \left[\mathbf{1}_{\{j=i\}} \alpha_i + \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \gamma_{ij} - \delta_{ij} \right] \\ & - \sum_{i=1}^n \sum_{j=1}^n M g_{ij} \gamma_{ij} + \sum_{i=1}^n \sum_{j=1}^n g_{ij} \delta_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} - \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj}. \end{aligned}$$

This dual formulation

4 Dismantlable Graphs

We expand Proposition 3 by considering all dominated vertices rather than just pendant vertices.

Theorem 1. *Singular triangular decomposition completely categorizes all dismantlable roots in the polynomial time linear program.*

References

- [RH60] Ian C. Ross and Frank Harary. “The square of a tree”. In: *The Bell System Technical Journal* 39.3 (1960), pp. 641–647.

A Result Diagrams

A.1 Linear Program

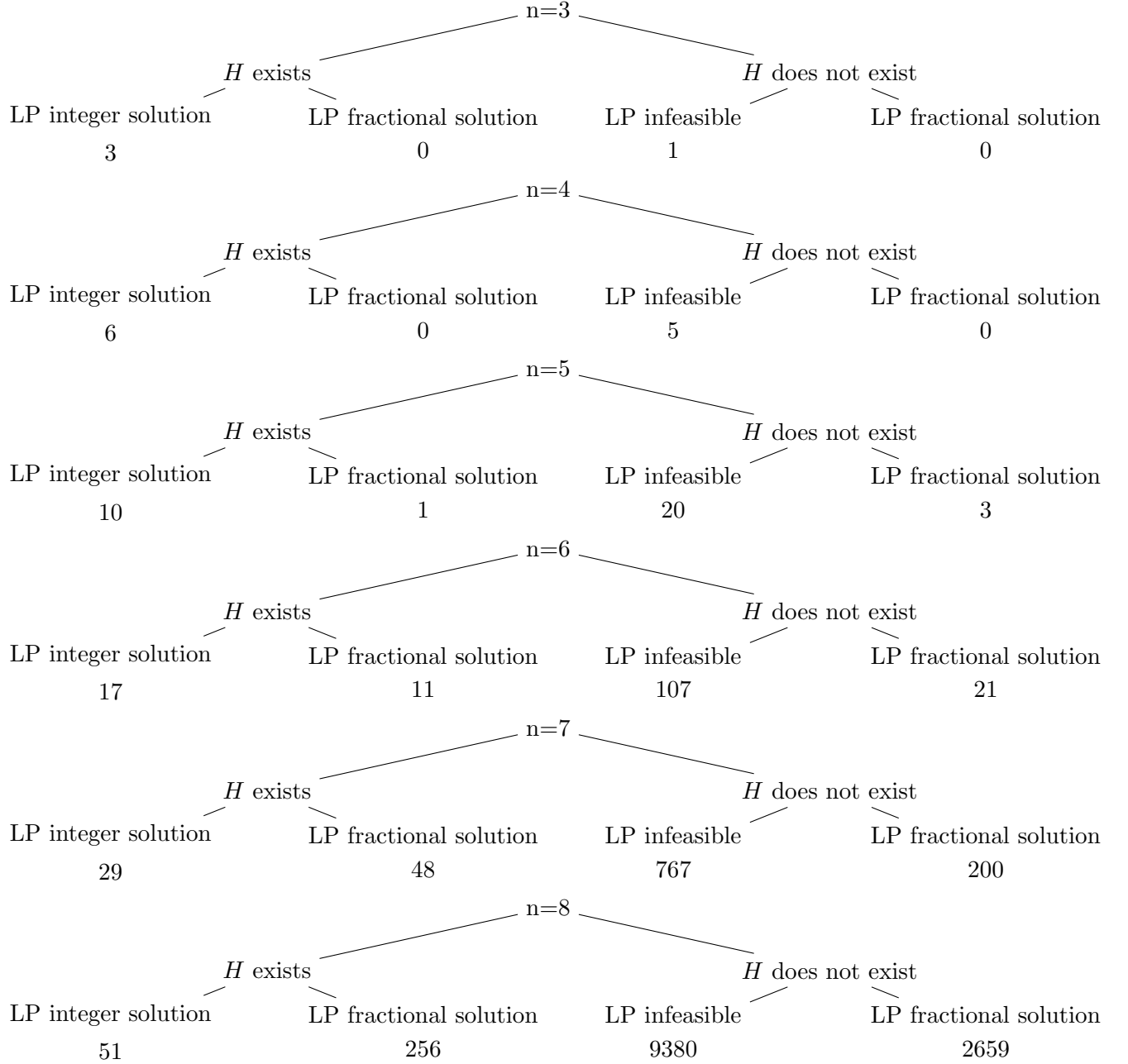


Figure 1: LP outcomes for different values of n on base linear program.

1

¹The total number of simple graphs on $n = 2, 3, 4, \dots$ unlabeled nodes is the sequence 2, 4, 11, 34, 156, 1044, 12346, $\dots$ given by A000088.

The IP integer solution sequence 2, 3, 6, 10, 17, 29, 51, $\dots$ is given by A285665.

The total number of graphs which are squares is the sequence 2, 3, 6, 11, 28, 77, 307, $\dots$ given by A382181.

A.1.1 Max Objective Function

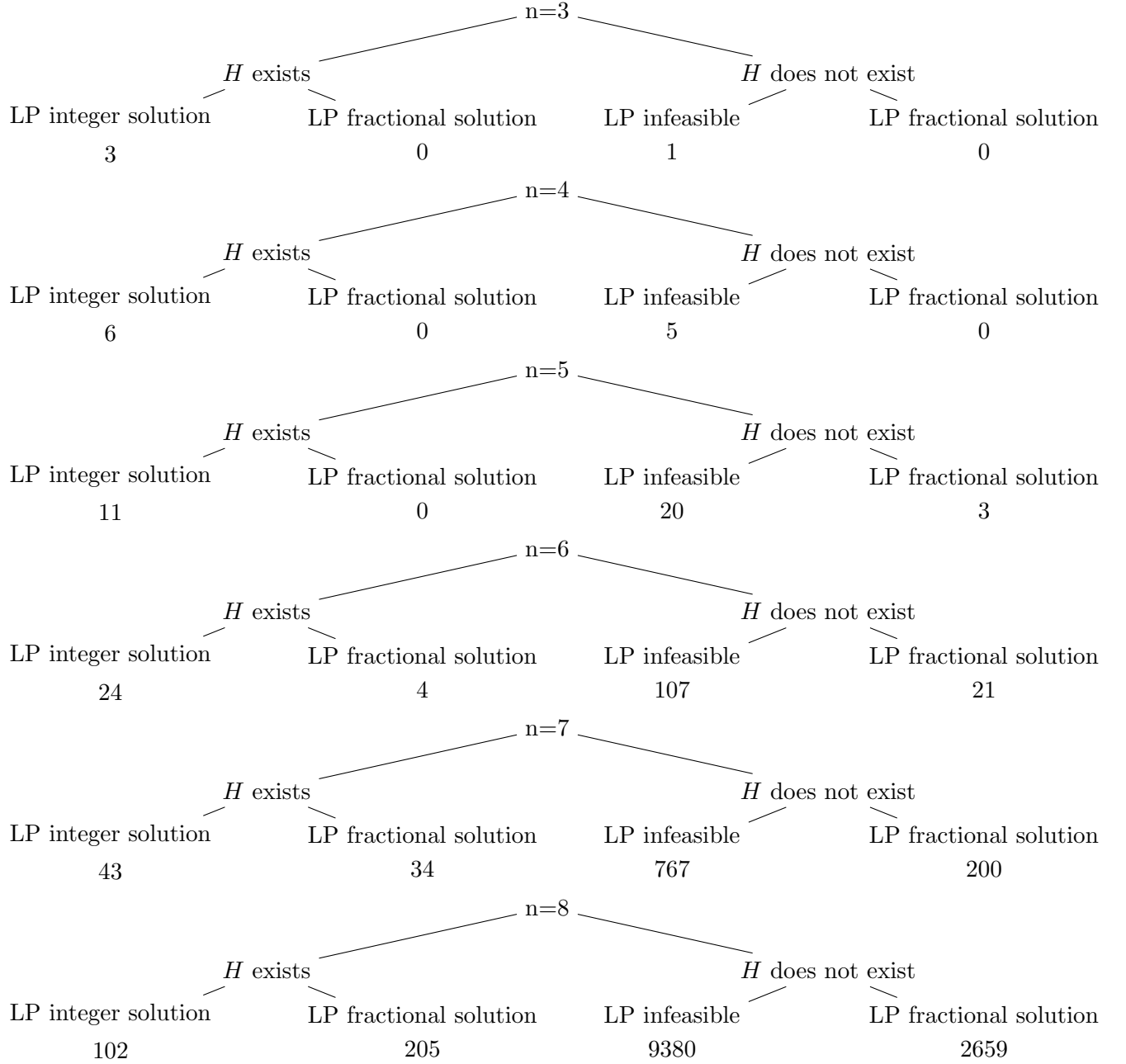


Figure 2: LP outcomes for different values of n on the linear program with max objective function.

2

²The LP integer solution sequence 3, 6, 11, 22, 42, 102, ... is very similar to the number of distinct characteristic polynomials of trees with n nodes - A226594.

A.1.2 Singular Triangular Decomposition

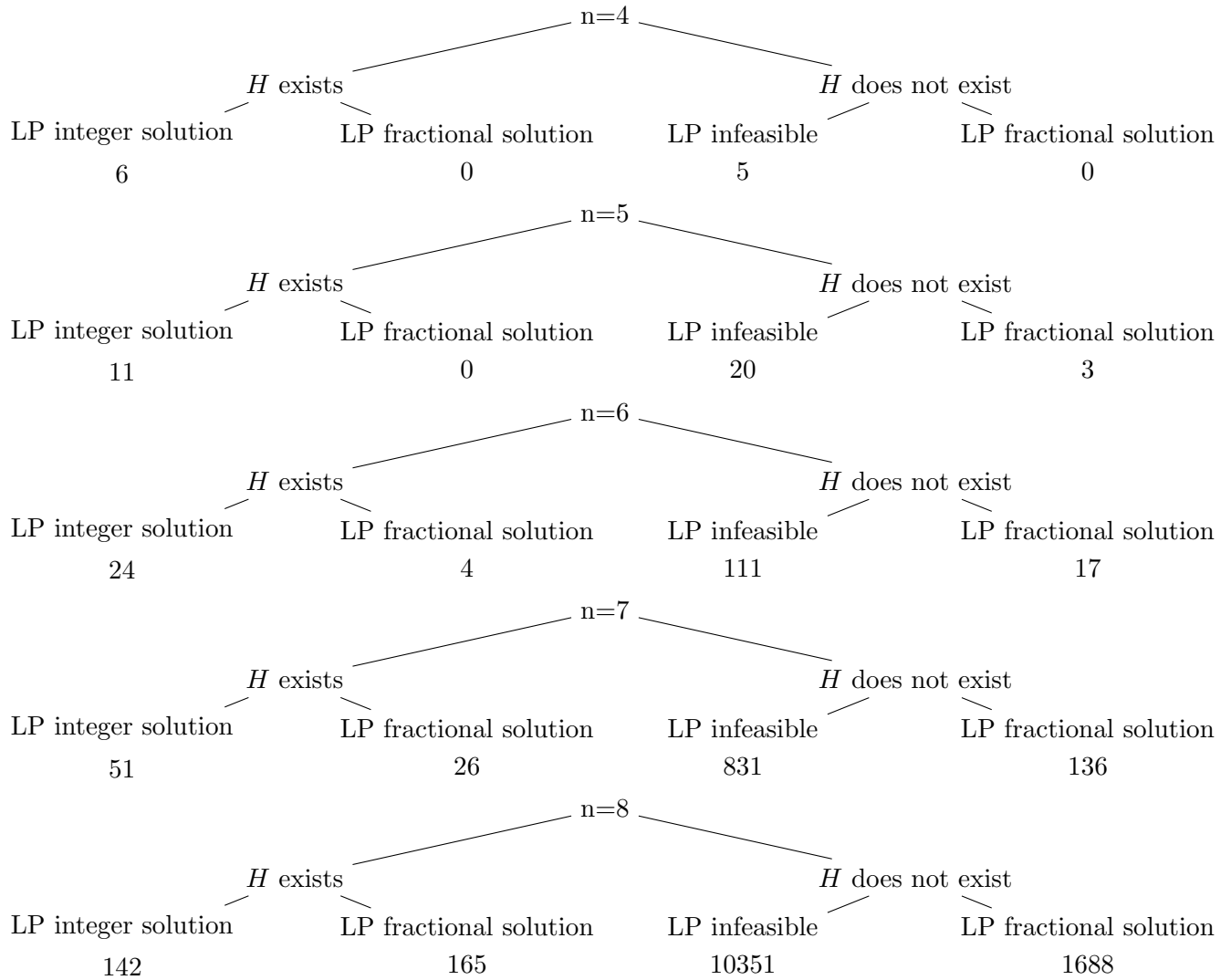


Figure 3: LP outcomes for different values of n on the linear program with max objective function and singular triangular decomposition.

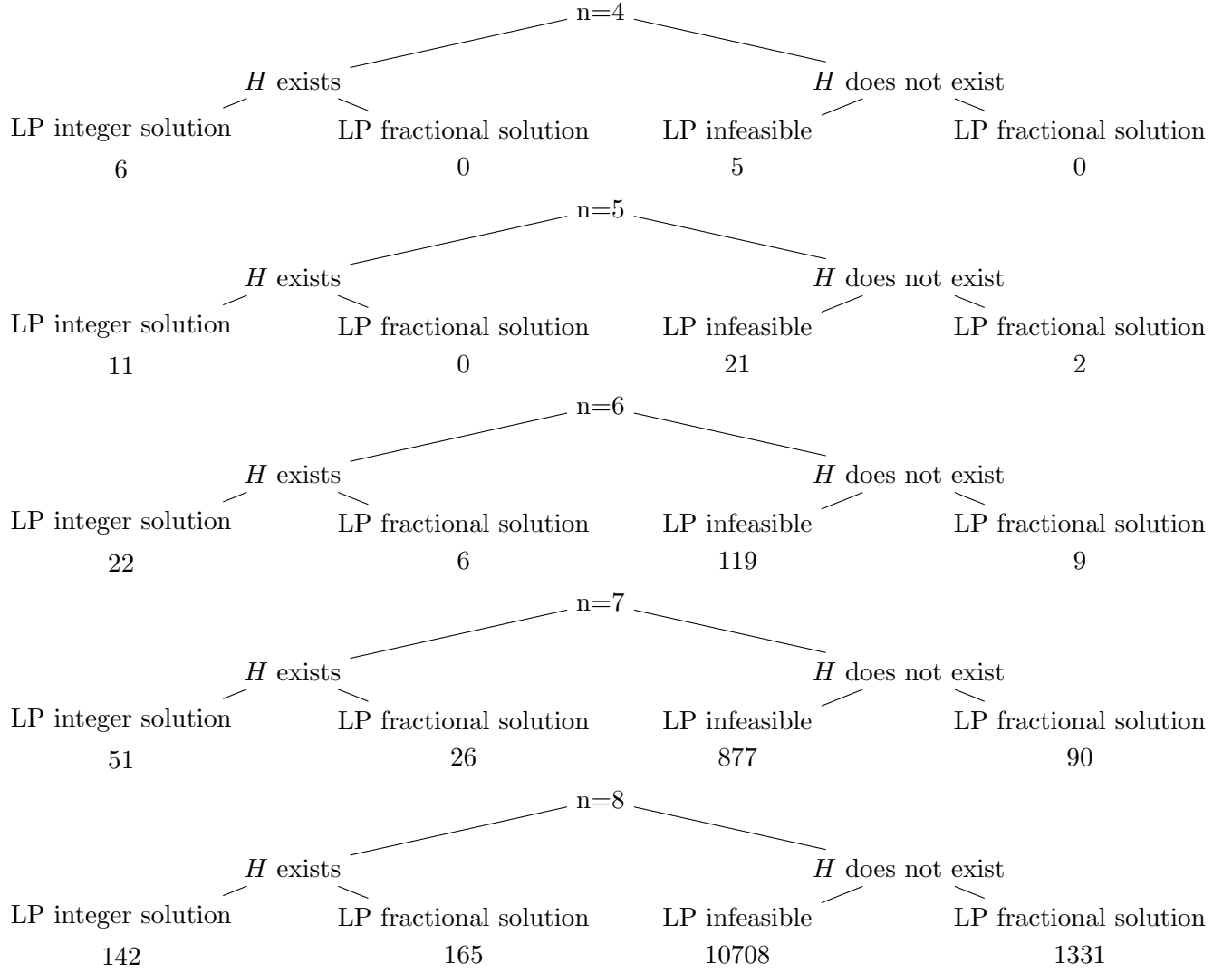


Figure 4: LP outcomes for different values of n on the linear program with max objective function and singular triangular decomposition with recursion for tie-breaking.

A.1.3 Rounding

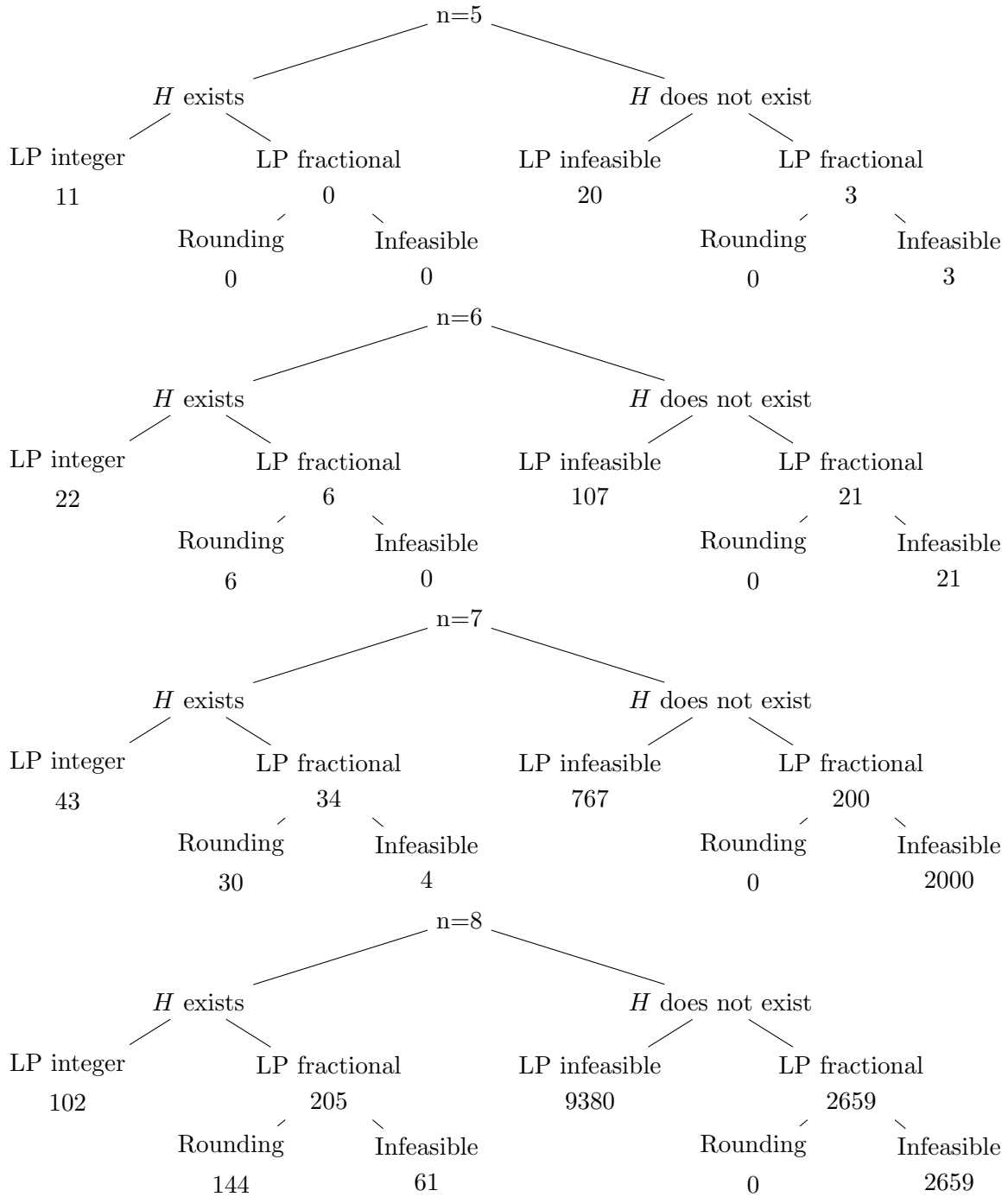


Figure 5: LP outcomes for different values of n on the linear program with max objective function and using rounding for fractional solutions.

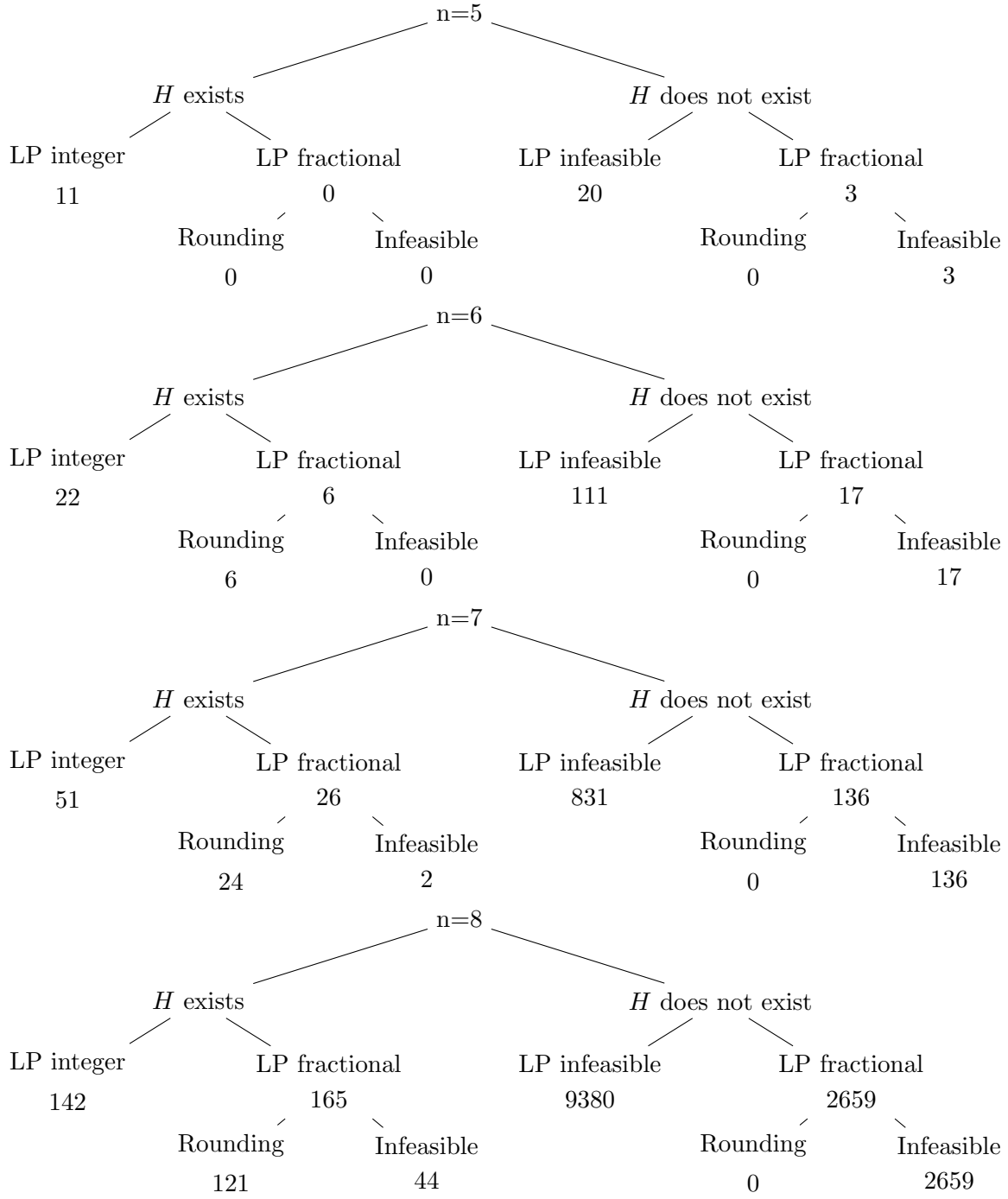


Figure 6: LP outcomes for different values of n on the linear program with max objective function, singular triangular decomposition and using rounding for fractional solutions.

A.2 Dual Program

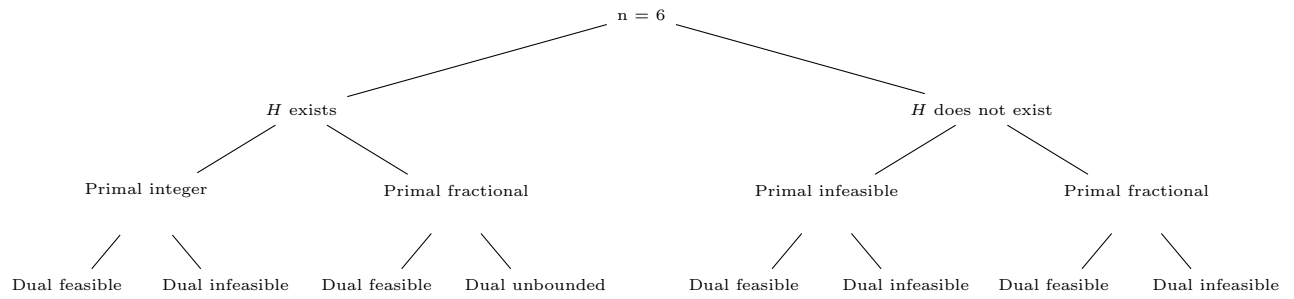


Figure 7: LP outcomes for different values of n on the dual linear program.